Thermodynamic Inference — Module 06: Learning — Discussion Questions

> Mix of analytical and applied questions for materials scientists and engineers. Questions 1-10 are analytical; Questions 11-20 are applied.

Analytical Questions

1. How does the Active Inference concept of learning map onto the physical processes observed in phase equilibria, transformation kinetics, and calphad? Identify at least three specific correspondences.

2. In what sense does a crystal lattice "minimize free energy" in the same way that an Active Inference agent minimizes variational free energy? Where does the analogy break down?

3. Compare the classical thermodynamic view of learning with the Active Inference interpretation. What additional insights does the FEP framework provide?

4. How does the concept of a Markov blanket help formalize the boundary conditions relevant to learning in metallurgical systems?

5. Explain how prediction error manifests in the context of learning during a non-equilibrium process such as rapid quenching of an Fe-C alloy.

6. How does the precision (inverse variance) of characterization measurements affect our ability to study learning in real materials?

7. Discuss the role of nested systems (atoms -> grains -> components) in the context of learning. How does information flow between scales?

8. What is the "generative model" that governs learning in a binary alloy system? How is this model encoded physically?

9. Compare how learning operates in a diffusion-controlled transformation versus a martensitic (diffusionless) transformation. What does this difference reveal about the "speed of inference" in materials?

10. How does the concept of epistemic value (information gain) apply to experimental design for studying learning?

Applied Questions

1. You are characterizing a new Ti-6Al-4V component after additive manufacturing. Design an experimental protocol that maximizes the epistemic value (information gain) regarding learning in this material.

2. A heat treatment schedule for a Ni-based superalloy is producing inconsistent results. Using the Active Inference framework, diagnose where the "prediction error" is likely originating and propose corrective actions.

3. Map the Markov blanket of a single austenite grain during the eutectoid transformation. What are the sensory states, active states, and internal states?

4. Design a simple computational experiment (using Python/PyCalphad) to demonstrate learning in the Fe-C system. Describe the expected outputs.

5. How would you use the concept of learning to improve quality control in a continuous casting process? Propose specific sensor placements and data analysis strategies.

6. A grain boundary in a polycrystalline copper sample is migrating under annealing. Describe this process in Active Inference terms: what is the agent, what is it "predicting," and what action is it taking?

7. Compare the "learning" processes of two alloy development approaches: traditional trial-and-error versus CALPHAD-guided design. Frame both in terms of learning and model updating.

8. You are tasked with designing a new high-entropy alloy. How does expected free energy (balancing pragmatic and epistemic value) guide your experimental plan for studying learning?
9. Describe how digital twin technology implements learning in an industrial heat treatment furnace. What constitutes the twin's generative model, and how is prediction error minimized?
10. Reflect on a real manufacturing problem you have encountered (or a published case study). Reinterpret the problem through the lens of learning in Active Inference. What new insights does this framing provide?